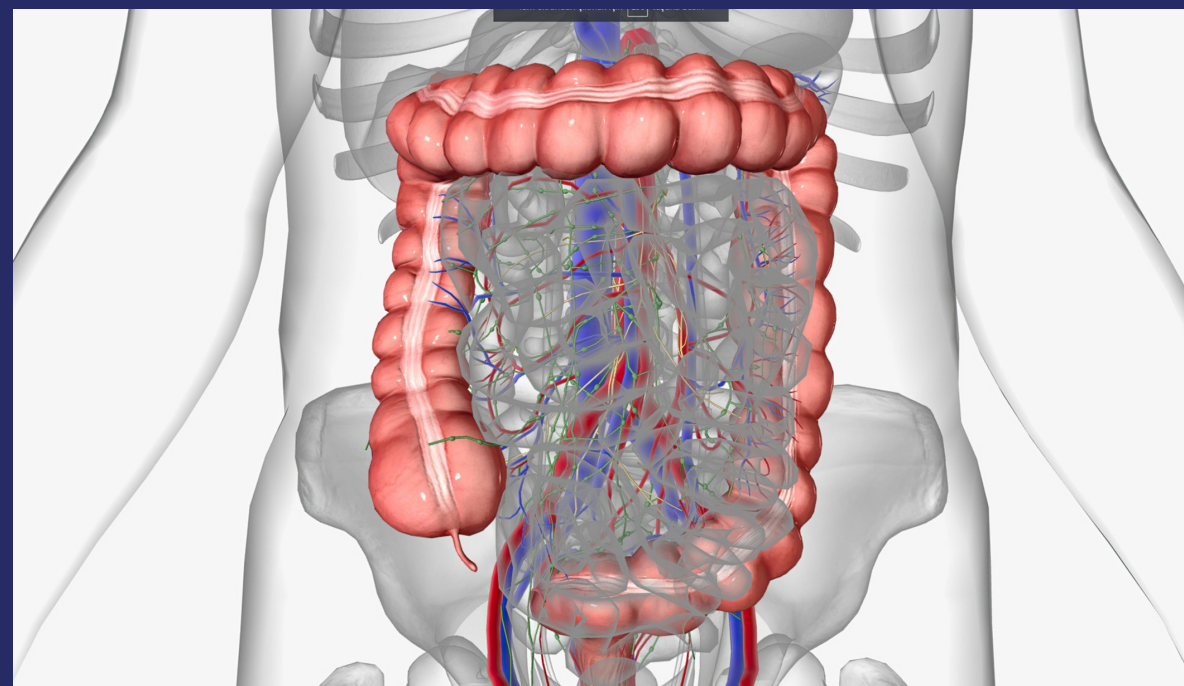


This work focuses on the use of "gellan gum" as a natural polymer in the development of "floating in-situ gels" for "gastroretentive drug delivery systems (GRDDS)". These systems are administered as liquids and transform into gels upon contact with gastric fluids, enhancing gastric retention and enabling sustained drug release. Gellan gum forms gels through ionic crosslinking or temperature changes, making it suitable for controlled release formulations. The study outlines different floating systems—effervescent, non-effervescent, and volatile liquid-based—and evaluates them based on viscosity, gelation time, buoyancy, drug content, and release profiles. Gellan gum's applications span various routes, including oral, ocular, nasal, and topical delivery. It improves drug bioavailability, reduces dosing frequency, and allows site-specific delivery. This book also reviews related patents, marketed products, and therapeutic applications for drugs like ofloxacin, pantoprazole, and clarithromycin, confirming gellan gum's effectiveness in developing innovative, sustained-release formulations.



Gyati Shilakari Asthana  
Ganti Meghamala  
Abhay Asthana

**Dr. Gyati Shilakari Asthana:** Professor at Gokaraju Rangaraju College of Pharmacy with 21+ years' experience, 63+ publications, 7 patents. Guided 3 Ph.D. scholars. Internationally awarded researcher and mentor.  
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# Gallen gum as a tool for floating in situ gel- patent and application

A Comprehensive Guide to Gallen Gum Based In-Situ Gel Technology and Applications



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